

AMIRTHA VARSHINI ELAVARASU

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Third-year BS Data Science student at IIT Madras (CGPA 9.0) who ships production software for the programme office — most recently taking the IITM Course Planner from a single-programme tool to a multi-programme platform and owning the Electronic Systems and Aeronautics rollout through to release. Comfortable across Python, Java, JavaScript, SQL and REST API design; looking for a software development internship where I can own a feature from design to production.

EDUCATION

Indian Institute of Technology Madras

2024 – Present

B.S. in Data Science and Applications — Third year | CGPA: 9.0 / 10

Relevant coursework: Data Structures & Algorithms (Python), Programming Concepts using Java, Database Management Systems, Modern Application Development I & II (+ projects), System Commands (Linux/Bash), Computational Thinking.

Justice Basheer Ahmed Sayeed College for Women

2018 – 2021

B.Com — CGPA: 7.6 / 10

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, SQL, HTML/CSS, C/C++ (basic)

Backend: Flask, FastAPI, Django, Node.js, Express.js, NestJS, SQLAlchemy, TypeORM, REST API design, Redis

Frontend: Vue.js, React.js, Next.js, Bootstrap, Mantine UI, responsive layout, Chart.js

Data & Tools: PostgreSQL, MySQL, BigQuery, pytest, Git & GitHub, GitHub Actions, Docker, Linux/Bash, Google Apps Script, Vercel, Render, Railway

EXPERIENCE

Tech Intern — Operations & Software Development — BS Programme Team, IIT Madras Feb 2026 – Oct 2026

- Extended the IITM BS Course Planner — a Flask / PostgreSQL application used by online-degree students to plan terms against prerequisite and progression rules — from a Data Science-only tool into a multi-programme platform, and owned the Electronic Systems and Aeronautics & Space Technology rollout through to production.
- Introduced a single per-programme configuration module holding credit caps, exit tiers, progression checkpoints and permitted programme switches, so course-availability, planner and dashboard logic stopped hard-coding one degree's rules; backed it with 18 ES/AE test modules inside a 190-test pytest suite.
- Mapped BigQuery warehouse datasets to programme codes and repaired the email-domain migration path so ES and AE students resolved to the correct account on first login; wrote a data-safe schema-normalisation migration (surrogate primary key, indexed account_id) using stub inserts in place of orphan deletes.
- Built Django applications that automate exam eligibility processing — category-wise pass marks and grace-mark rules previously applied by hand across TSV mark sheets — and a FastAPI service on Railway that generates travel-allowance PDFs from Google Forms submissions.
- Work through peer code review with the programme team and maintain the NPTEL Local Chapter database; a joint pre-release audit before the ES/AE launch caught seven crash and authentication defects before they reached students.

Front-End Development Intern — Edunet Foundation (AICTE & IBM SkillsBuild)

Aug 2025 – Sep 2025

- Completed a 6-week programme building responsive web interfaces in HTML, CSS and JavaScript, working to industry review and delivery workflows.

Junior Accounts Executive — Veranda Race, Chennai

Nov 2021 – Mar 2023

- Automated recurring reconciliation and reporting workflows, cutting manual effort in the monthly close and reducing data-entry errors across branch submissions.

PROJECTS

Quiz Master v2 — Flask-RESTful, SQLAlchemy, Vue.js, Bootstrap, Chart.js

- Designed an API-first quiz platform with role-based access for admins and students, timed attempts with automatic evaluation on submit, and Chart.js analytics summarising per-subject performance; REST endpoints are consumed by a Vue single-page app.

Dynamic Portfolio — Flask, Vue 3 + Vite, JWT

- Built a portfolio application with JWT authentication and an admin CRUD interface for editing content without a redeploy, structured around composables, a client-side router and reusable components.

LEADERSHIP & ACTIVITIES

Event Head, RoboSoccer 5.0 — Paradox '26, IIT Madras

- Led planning and execution for the RoboSoccer event across a volunteer team: sponsorship outreach (NukePC, HCL GUVI, Karthikesh Robotics), volunteer recruitment, and workshops on ESP32/Pico microcontrollers and motor drivers.
- Built a WiFi-controlled robot on ESP32 with L298N drivers and WebSocket control, and an FPV RC car on XIAO ESP32S3 Sense with NRF24L01 radio.